

IF 45 I

Advance Web Programming

Meet I - Introduction to Node.js & Environment Setup

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Syllabus

1. Introduction to Node.js & Environment Setup
2. Modules & Packages in Node.js
3. JSON Basics & CRUD with Files
4. Mini Project: Student Management (JSON)
5. Introduction to MySQL Database
6. Connecting Node.js to MySQL Database
7. CRUD Operations in MySQL Database
8. Introduction to Express.js
9. Middleware & Routing in Express
10. REST API with Express + MySQL Database
11. Connecting Express API to ReactJS
12. File Upload & Basic Authentication
13. Fullstack Mini Project Case Study
14. Final Project Presentation

Evaluation

- Midterm Exam (UTS) : 30%
- Final Exam (UAS) - Project Based : 40%
- Activity: 30%
 - ✓ Quiz
 - ✓ Daily challenges
 - ✓ Student activity

Mid Project Rules

- Maximum person in Group 3-4 persons
- Your website must be published in website hosting
- Your website project can integrate with other subject (such as Database design, Mobile Application Programming, Cybersecurity)

Meet I

Objective:
Sub-CLO07 I I

Students are able to explain the basic concepts of web development.
(C2)

OUTLINE

- Backend vs frontend.
- Role of Node.js in web development.
- Installing Node.js, npm, and VSCode.
- Running JavaScript in Node.js.

Backend vs Frontend

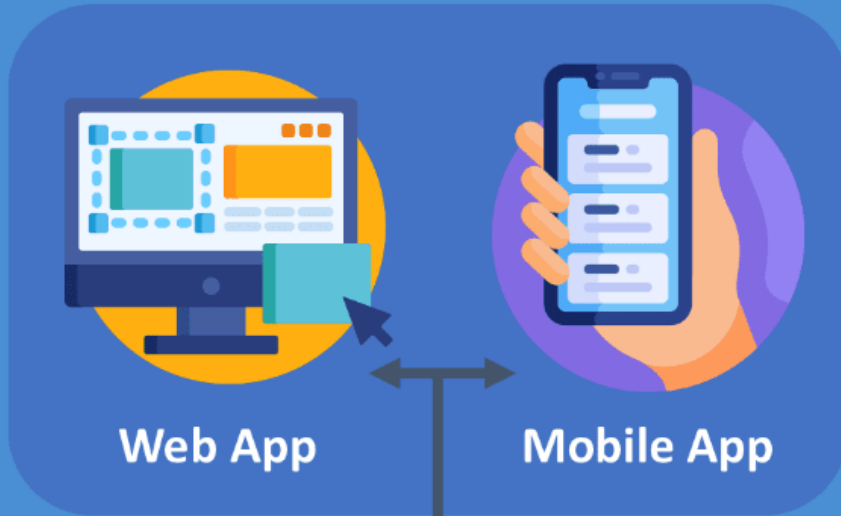
Frontend

- The frontend is the part of a software application that users can see and interact with directly.
- It deals with the user interface (UI) and the user experience (UX).
- It is built using technologies like HTML, CSS, and JavaScript (and frameworks like React, Angular, Vue).
- The main purpose is to display data and provide an easy way for users to interact with the system.

Backend

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FRONT-END



Web App

Mobile App

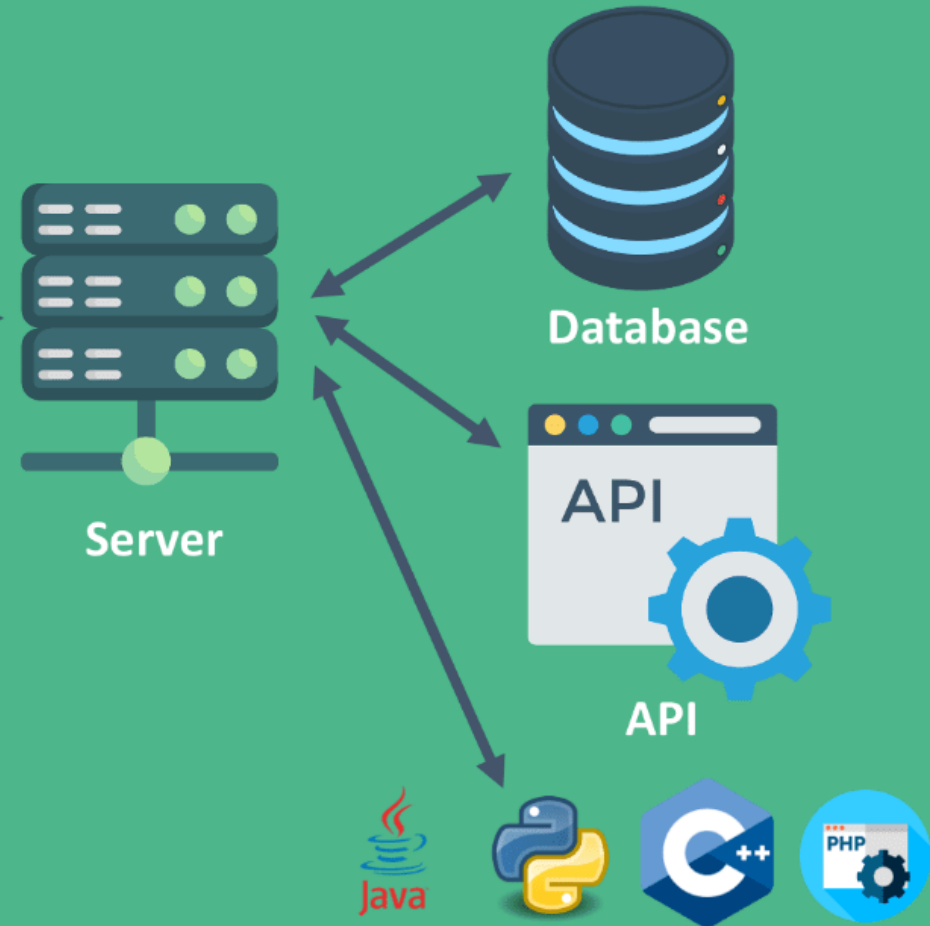
HTML

CSS

JS



BACK-END



Server

Database

API

API



Introduction

- Official website: <https://nodejs.org/>
- Visual Studio code (or Notepad++) or other Text Editor

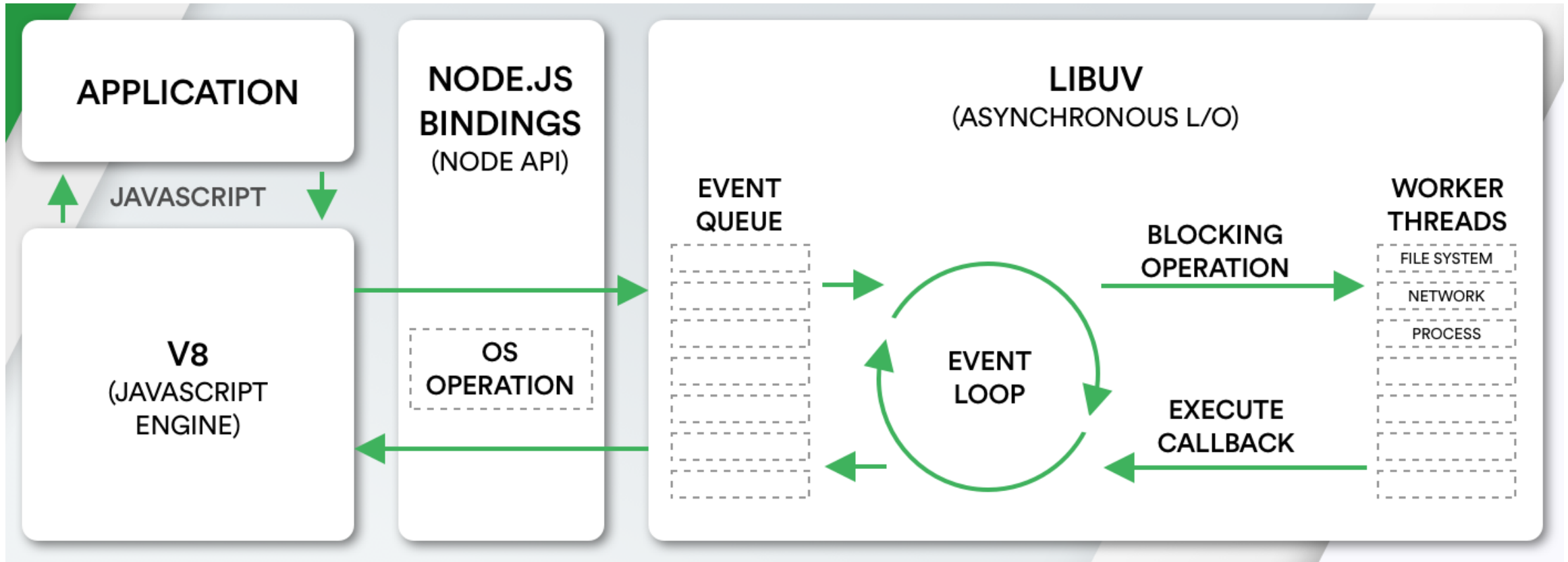
What is NodeJS?

Node.js is an open-source and cross-platform JavaScript runtime environment. It is a popular tool for almost any kind of project

Node.js runs the V8 JavaScript engine, the core of Google Chrome, outside of the browser. This allows Node.js to be very performant.

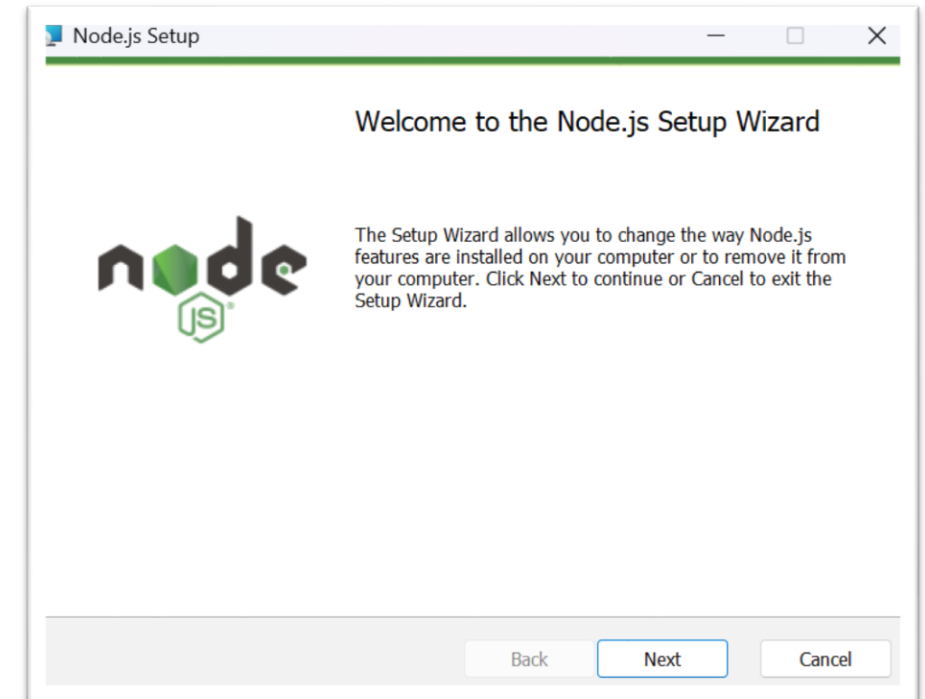
A Node.js app runs in a single process, without creating a new thread for every request. Node.js provides a set of asynchronous I/O primitives in its standard library that prevent JavaScript code from blocking. In addition, libraries in Node.js are generally written using non-blocking paradigms. Accordingly, blocking behavior is the exception rather than the norm in Node.js.

NodeJS Architecture



Node Installation Instructions

1. Visit <https://nodejs.org/en/download>
2. Download NodeJS versi LTS (Windows Installer (.msi))
3. Install NodeJS to your computer



Node Command

1. Open your command prompt (cmd)
2. To test your node, type
node -v atau **node --version**
3. To find out which versions of Node.js you may have installed, type
npm ls
4. To change your nodeJS version, type:
npm use codeversion
Sample: **npm use 22.13.1**

Npm install

When you install Node.js from the official website (nodejs.org), npm (Node Package Manager) is automatically installed.

To check npm installation in command prompt:

```
npm -v
```

Create folder and file

1. To create new project, create new folder and rename it to “**sample_project**”
2. Create new file in the your project folder and rename it to “**app.js**”
3. Write this code in your file
4. Open browser, dan type : **http://localhost:3000**

```
// Import module bawaan Node untuk HTTP
const http = require("http");

// Buat server
const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/plain" });
  res.end("Hello World from Node.js asdasdasd!\n");
});

// Jalankan server di port 3000
server.listen(3000, () => {
  console.log("Server berjalan di http://localhost:3000");
});
```

Manual Reload

If you change your code, like the sample below

```
// Buat server
const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/plain" });
  res.end("Hello World edit by Wirawan!\n");
});
```

Note:

1. After change code, you should stop your server, with press **ctrl+c** in command prompt
2. Run your server again, with command: **node app.js**
3. Refresh your browser

Nodemon

Nodemon is a tool that helps develop Node.js based applications by automatically restarting the node application when file changes in the directory are detected.

nodemon does not require any additional changes to your code or method of development. nodemon is a replacement wrapper for node. To use nodemon, replace the word node on the command line when executing your script.

Official website: <https://www.npmjs.com/package/nodemon>

Install Nodemon

Install Nodemon

npm install -g nodemon

Run server using Nodemon

nodemon app.js

atau

npx nodemon app.js

Note: After install Nodemon, you no need to stop and start server node anymore, you just only need to refresh your browser

NEXT WEEK'S OUTLINE

- Modules & Packages in Node.js

REFERENCES

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- R. Raj, Learning Node.js Development, Packt Publishing, 2018.
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- The Node.js Foundation, “Node.js Documentation,” [Online]. Available: <https://nodejs.org/en/docs/>. [Accessed: Jun. 5, 2025].
- Express.js Team, “Express.js Documentation,” [Online]. Available: <https://expressjs.com/>. [Accessed: Jun. 5, 2025].

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.